

MMF062 Vehicle Motion Engineering

Design Task 1 - Group 5

George Vasanth Fransua S

Ramkumar Huleppa Munavalli

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Task 1 - Tire Characteristics on different Road Conditions

Plot tyre model

The tyre model plotted using the 'Magic formula' is shown here using different parameters for different surfaces.

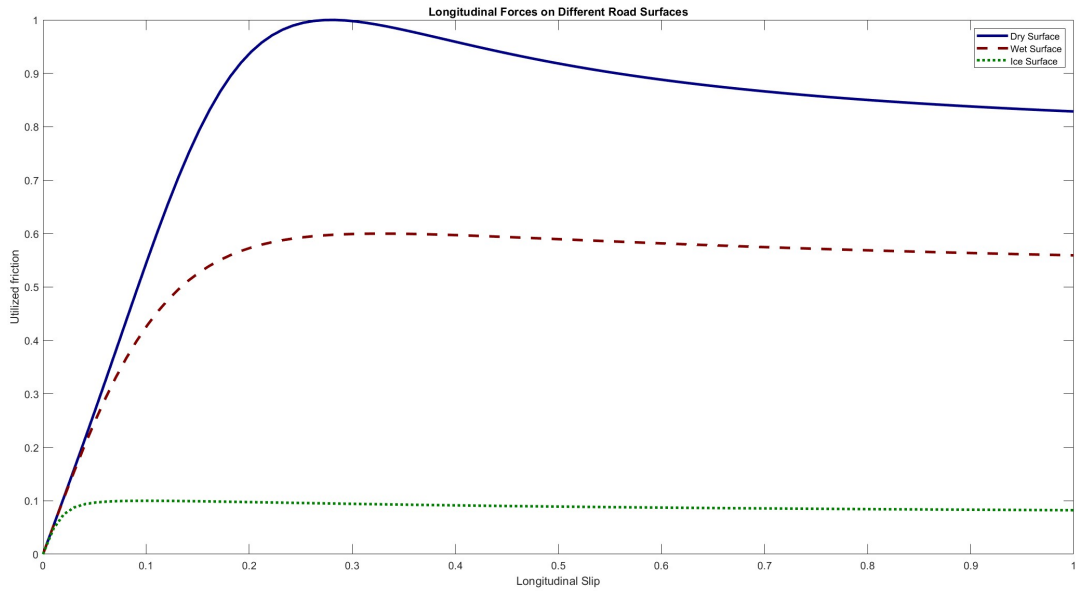


Figure 1: Normalised forces acting on different surfaces

Plot experimental tyre behaviour

The nominal parameters for C and D for the first experimental data when $E=0$ was found to be **2.75** and **0.94** respectively. Similarly, when $E=-4.0$ the values for C and D were **1.5** and **0.94** respectively. The plots are shown below

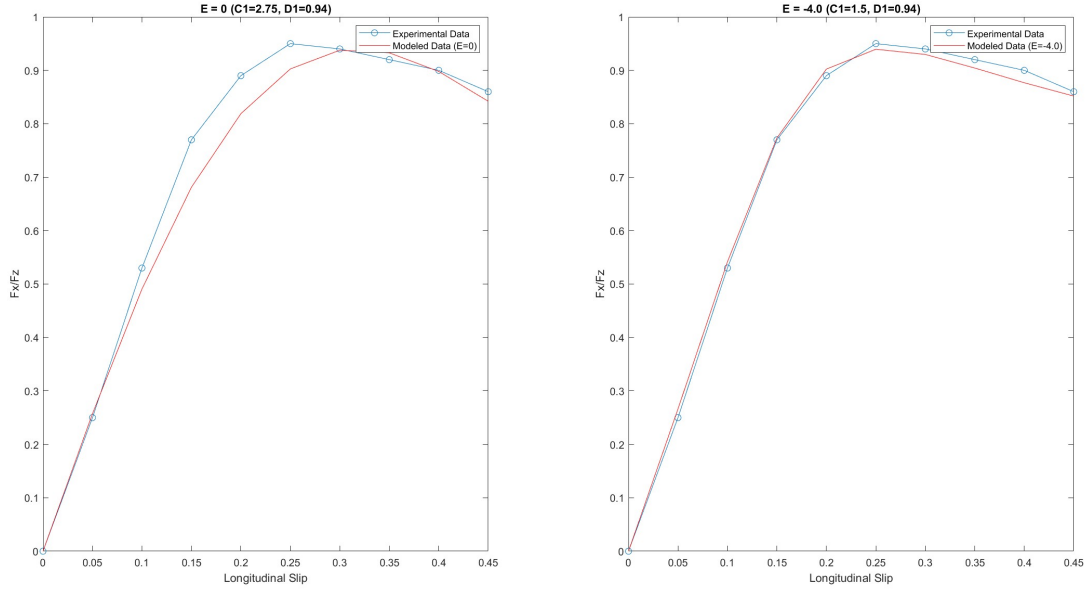


Figure 2: Experimental data 1

The nominal parameters for C and D for the second experimental data when $E=0$ was found to be **1.60** and **0.85** respectively. Similarly, when $E=-4.0$ the values for C and D were **1.55** and **0.99** respectively. The plots are shown below

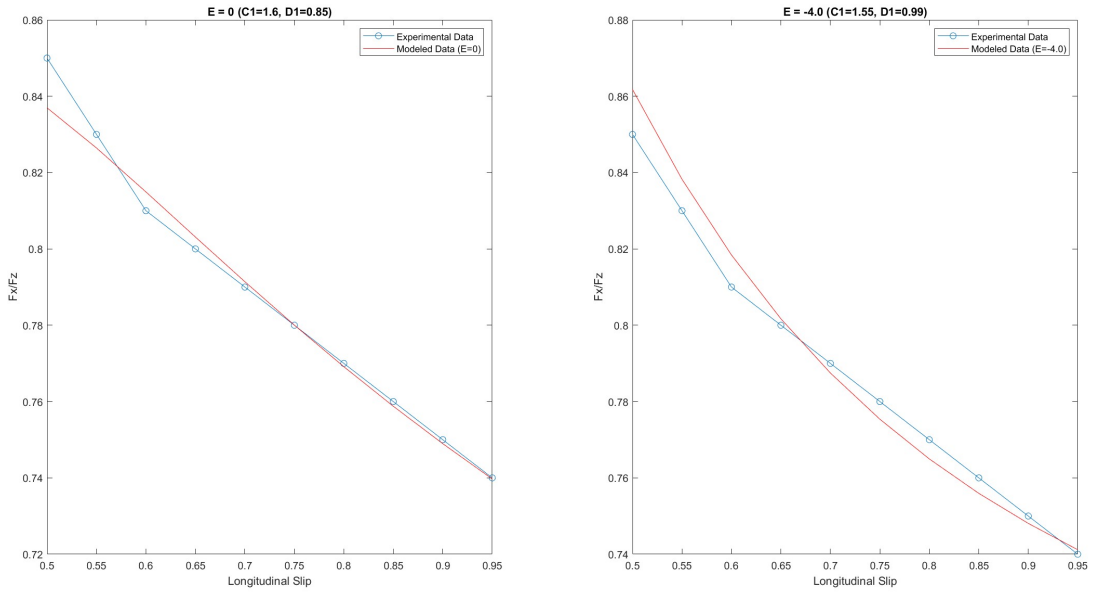


Figure 3: Experimental data 2

Task 2: Vehicle Model

Physical model (incl. free body diagram)

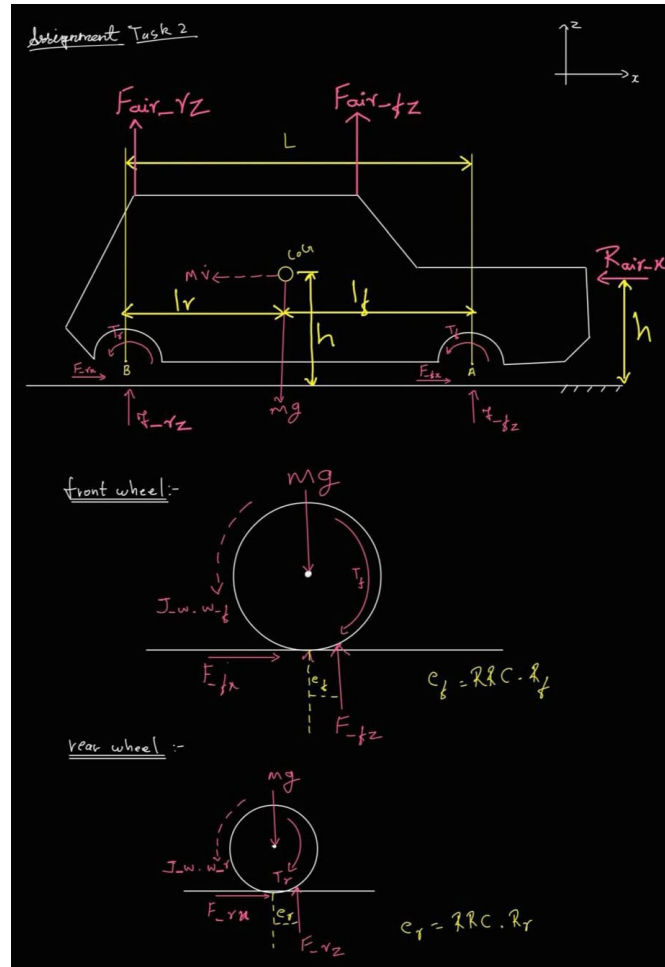


Figure 4: FBD

Mathematical model

When the vehicle is driving on the horizontal road (No slope)

1. Vehicle Longitudinal Force

$$M \cdot \dot{v} = F_{fx} + F_{rx} - R_{air} - M \cdot g \quad (1)$$

2. Normal Forces Acting on the Vehicle

$$F_{fz} + F_{air.fz} + F_{rz} + F_{air.rz} = M \cdot g \quad (2)$$

3. Moment About Point A (Front Wheel)

$$(F_{rz} + F_{air.rz}) \cdot L = M \cdot \dot{v} \cdot h + M \cdot g \cdot l_f + R_{air} \cdot h + M \cdot g \cdot h \quad (3)$$

4. Tractive Force on the Front and Rear Wheels

$$F_{fx} = S_{fx} \cdot F_{fz} \quad (4)$$

$$F_{rx} = S_{rx} \cdot F_{rz} \quad (5)$$

5. Wheel Rotational Forces

Front Wheel:

$$T_f - F_{fx} \cdot R_f - F_{fz} \cdot RRC \cdot R_f = J_w \cdot \omega_f \quad (6)$$

Rear Wheel:

$$T_r - F_{rx} \cdot R_r - F_{rz} \cdot RRC \cdot R_r = J_w \cdot \omega_r \quad (7)$$

Update models with road gradient

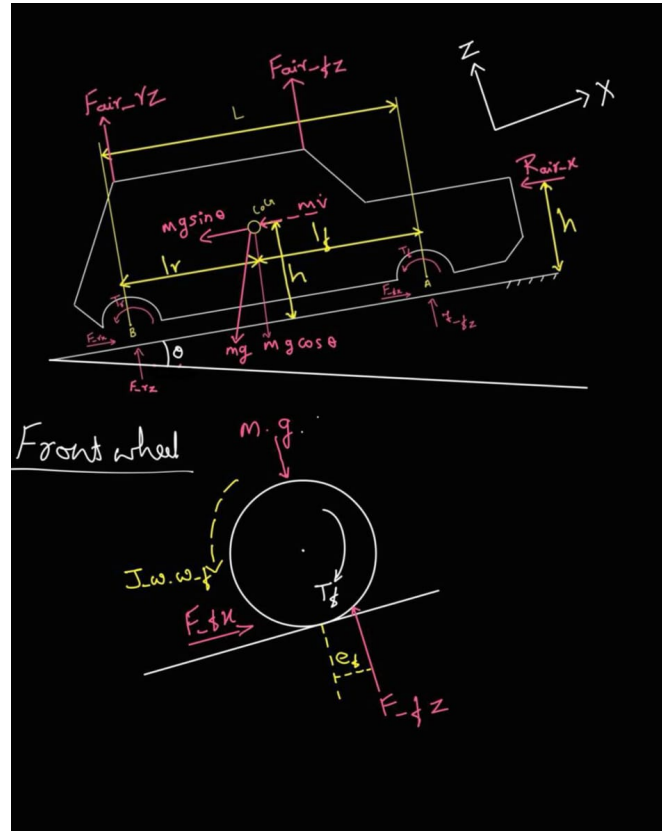


Figure 5: FBD with slope

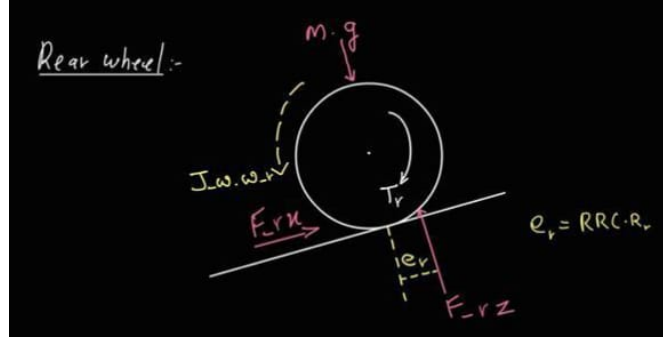


Figure 6: FBD with slope(Rear tyre)

Solve the equations

Vehicle Longitudinal Dynamics with Gradient

1. Vehicle Longitudinal Force

$$M \cdot \dot{v} = F_{fx} + F_{rx} - R_{air} - M \cdot g \cdot \sin(\theta) \quad (8)$$

2. Normal Forces Acting on the Vehicle

$$F_{fz} + F_{air.fz} + F_{rz} + F_{air.rz} = M \cdot g \cdot \cos(\theta) \quad (9)$$

3. Moment About Point A (Front Wheel)

$$(F_{rz} + F_{air.rz}) \cdot L = M \cdot \dot{v} \cdot h + M \cdot g \cdot \cos(\theta) \cdot l_f + R_{air} \cdot h + M \cdot g \cdot \sin(\theta) \cdot h \quad (10)$$

4. Tractive Force on the Front and Rear Wheels

$$F_{fx} = S_{fx} \cdot F_{fz} \quad (11)$$

$$F_{rx} = S_{rx} \cdot F_{rz} \quad (12)$$

5. Wheel Rotational Forces

Front Wheel:

$$T_f - F_{fx} \cdot R_f - F_{fz} \cdot RRC \cdot R_f = J_w \cdot \omega_f \quad (13)$$

Rear Wheel:

$$T_r - F_{rx} \cdot R_r - F_{rz} \cdot RRC \cdot R_r = J_w \cdot \omega_r \quad (14)$$

Task 3: Simulation of Longitudinal Dynamics

Code the slip definition in Matlab function

Please refer the code for the implementation.

Implement and validate model

The equations we got for **2d** have been implemented and the following plots were produced. The one which is shown right below is a scenario of AWD on a dry surface with a 8 degree inclination. Here we can see that the distance covered is the maximum amongst the other options (which were only either front wheel/rear wheel drive). Both the motor speeds seem to reach some optimal level for the given gear. Obviously the speed is going to be higher in this case since it's an all wheel drive. The slips are pretty similar although the rear is slipping lesser than the front and this is because the load on the rear side while climbing the hill is significantly higher so we tend to have better grip therefore lesser slip.

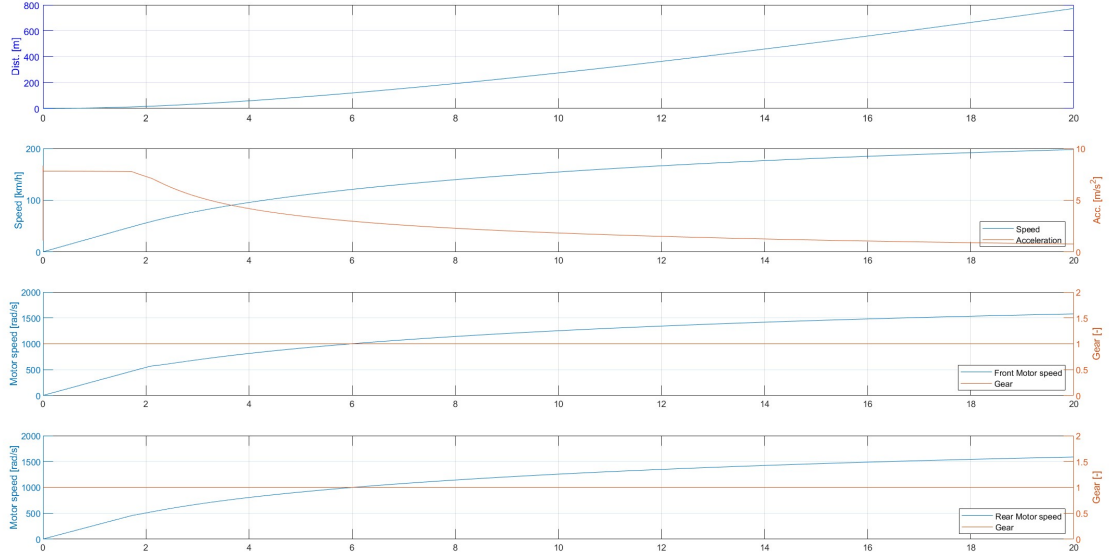


Figure 7: AWD on dry surface with 8 degree inclination

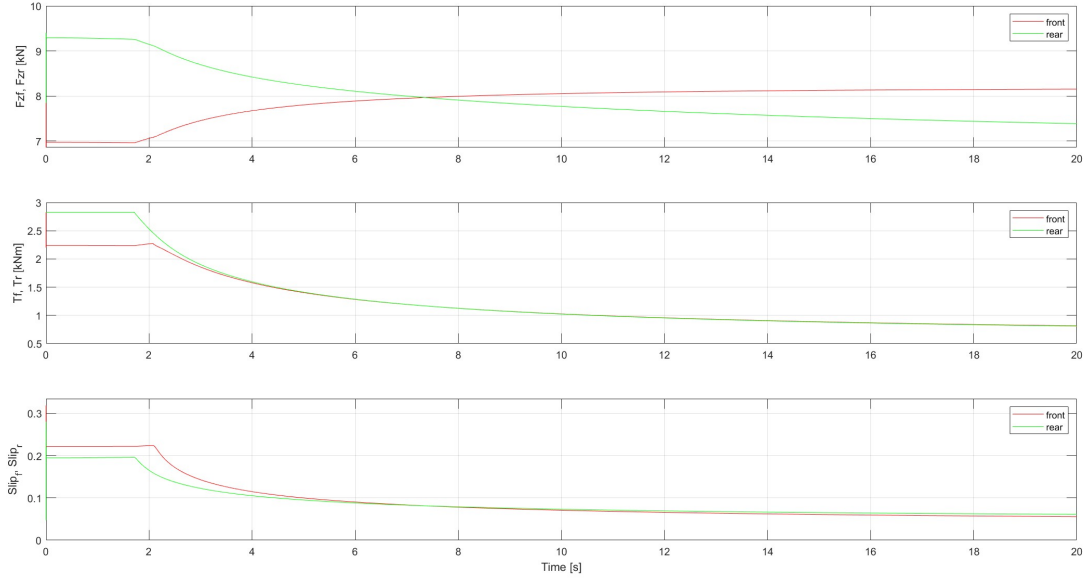


Figure 8: AWD on dry surface with 8 degree inclination - wheel specific plots

Did a similar test using only FWD now. Here we can see that the distance covered and the speed is going to be lesser than the AWD scenario for obvious reasons. The front wheel motor speed is going to be a bit higher than the rear. In this case only the front wheel is going to experience slip and torques. The plots are shown below.

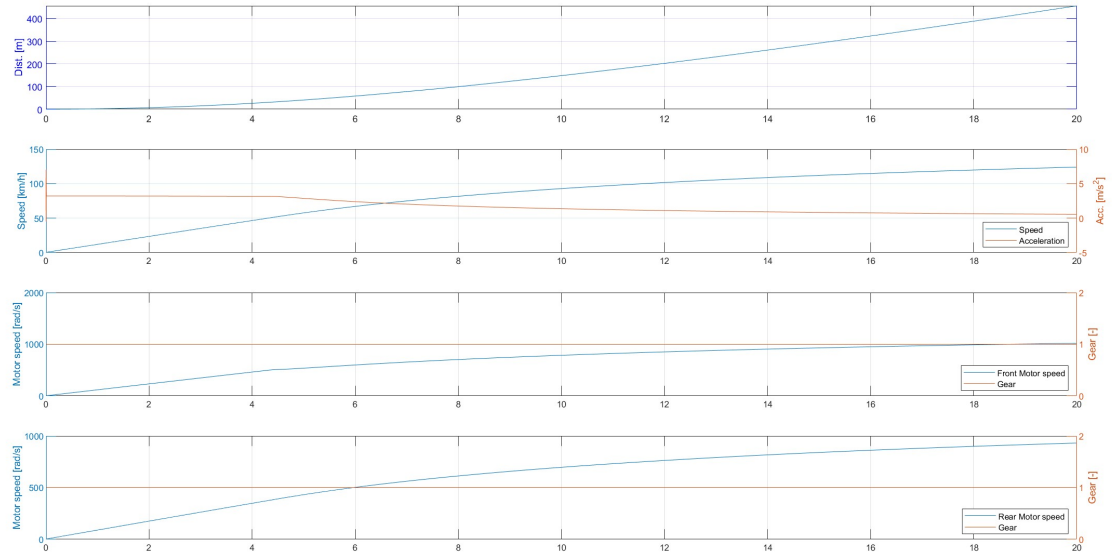


Figure 9: FWD on dry surface with 8 degree inclination

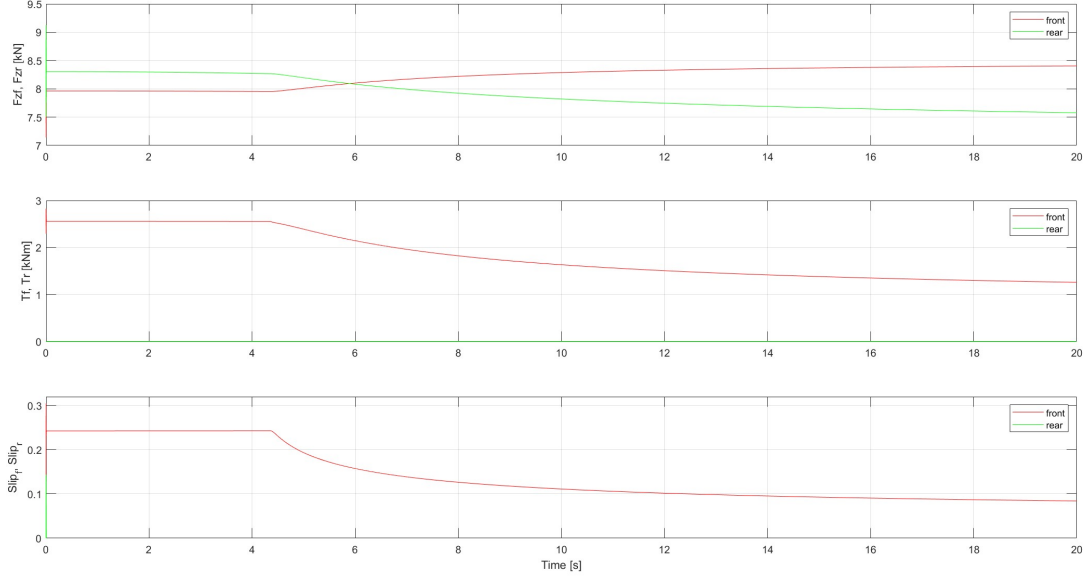


Figure 10: FWD on dry surface with 8 degree inclination - wheel specific plots

The similar test is done on the RWD. We can see that the distance covered and the speed is going to be same as the FWD. The rear wheel motor speed is going to be a bit higher than the front wheel motor speed and rear wheel will be experiencing the slip and the torques.

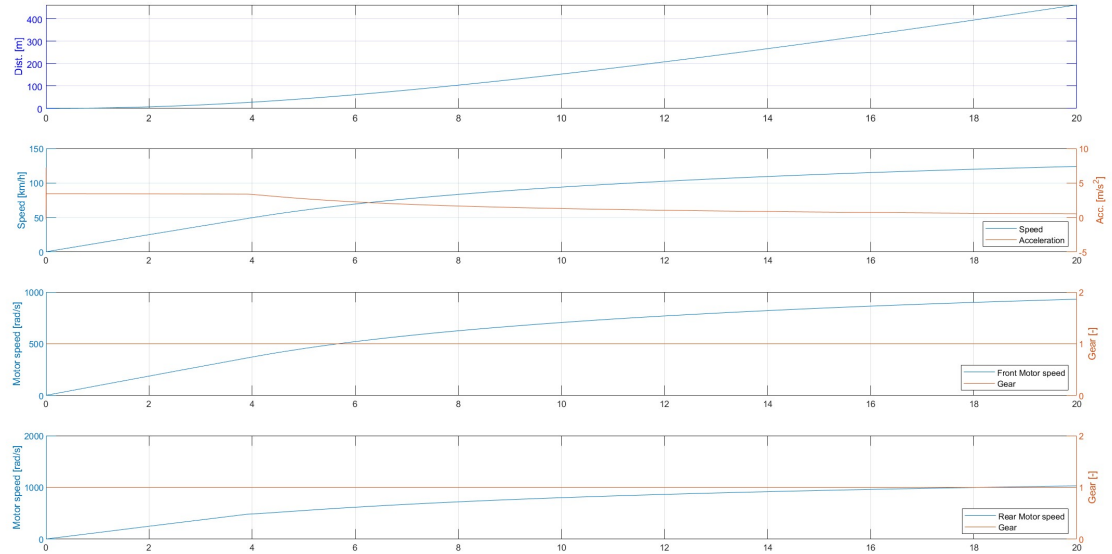


Figure 11: RWD on dry surface with 8 degree inclination

From these plots we can reasonably conclude that the dynamic equations are correct.

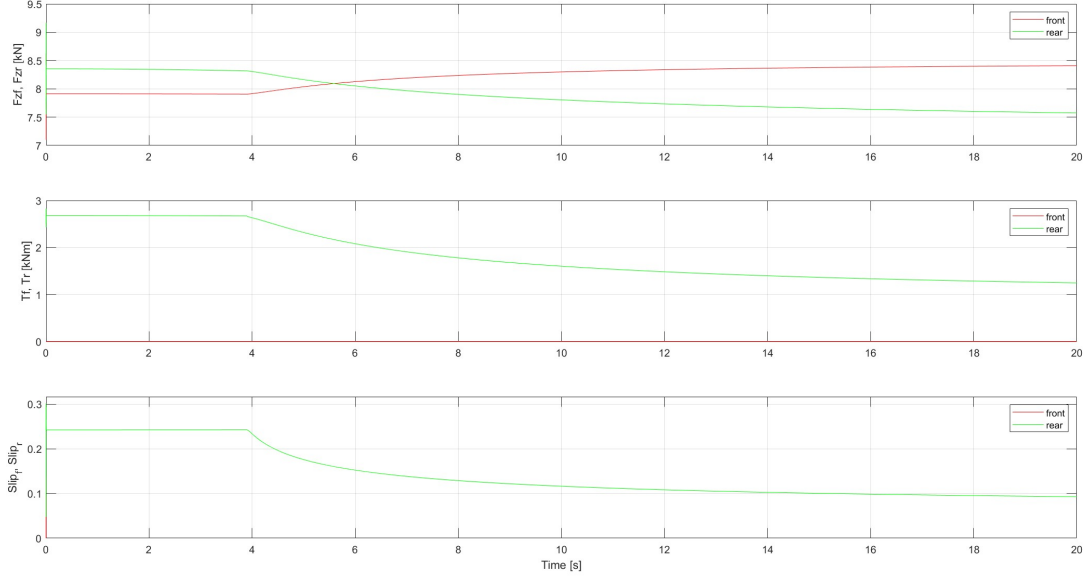


Figure 12: RWD on dry surface with 8 degree inclination- wheel specific plots

Select between FWD and RWD

Drive mode	Dry	Wet
RWD	7.823 s	12.179 s
FWD	8.025 s	11.532 s

Table 1: Timings on different drive modes for 100m

In my opinion since we know that there is a high chance of rain I would choose the **FWD** setup as it gives a lesser time by 0.65 seconds relative to the RWD setup. Although the RWD performs better when it's dry by 0.2 seconds it is safer to say that the FWD is better suited in this scenario cause the percentage gain is more for the wet surface scenario.

Task 4: Vehicle Model

Power consumption of vehicles with TC

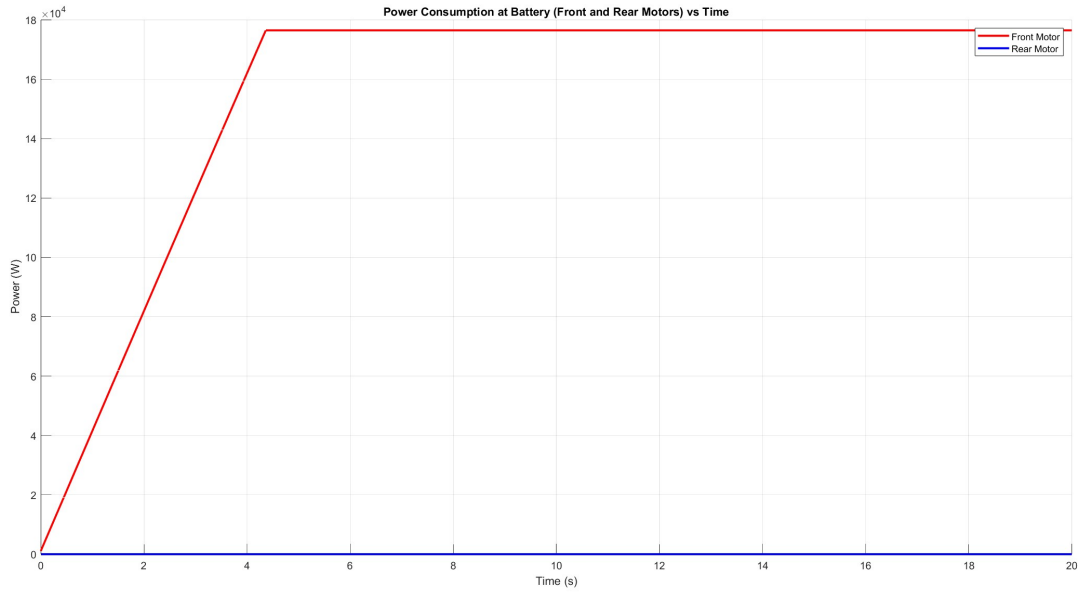


Figure 13: FWD on dry surface

For FWD vehicles, the traction is experienced only on the front wheels which is represented in red in the above graph. The transient response for it is about 4.2 sec which is more compared to the RWD shown in from fig 15. This is because we have an inclination and going upwards so the normal forces acting on the rear side is more and thus we get more traction thereby reaching the power limit with lesser transient time.

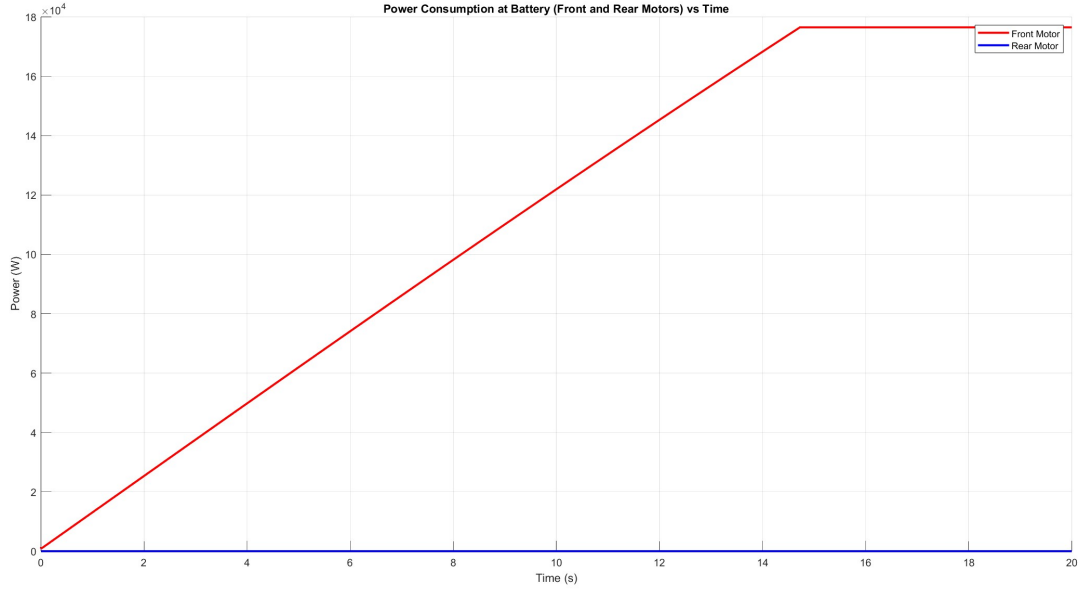


Figure 14: FWD on wet surface

On the contrary in a wet surface the transient time is more which is about 14.7 sec due to lack of traction and slippage but since we have traction control enabled it eventually gets full traction. This is shown in the graph in a red line.

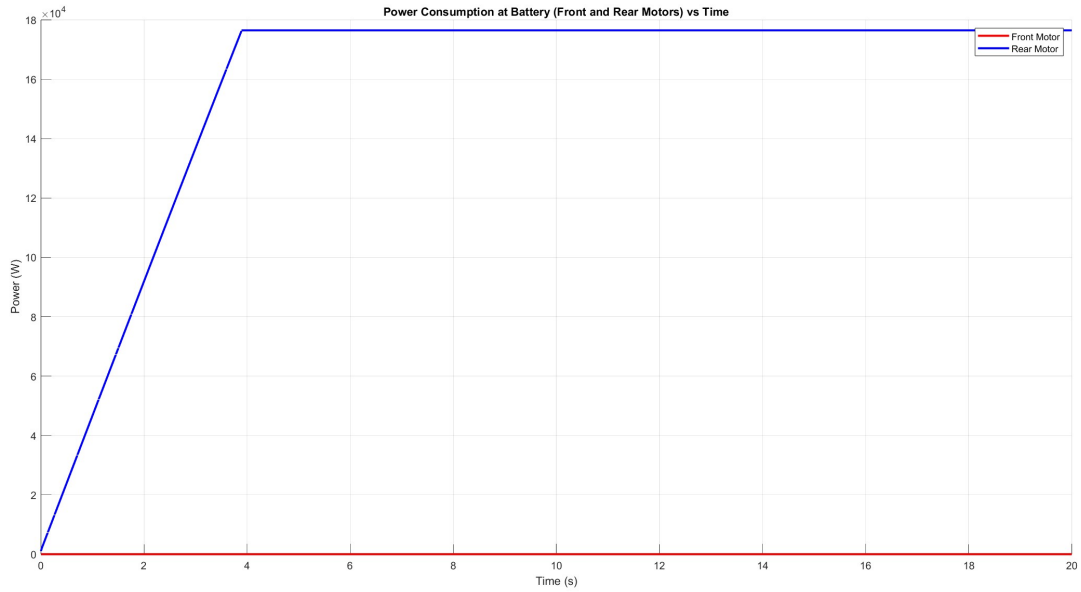


Figure 15: RWD on dry surface

For RWD vehicles the propulsion is take care by the rear axle as the power is delivered to the rear wheels (blue lines). The reasons explaining why it has lesser transient time is explained on the FWD dry surface scenario.

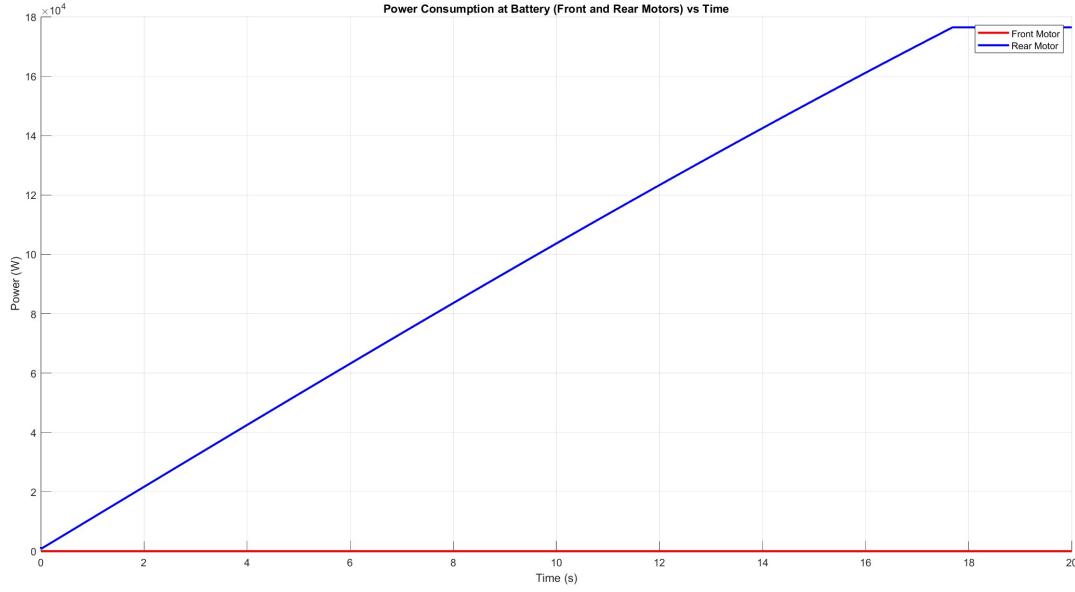


Figure 16: RWD on wet surface

From the above figure, we can see that the transient response of power consumption is about 17.6 sec on wet surface due to lack of traction of RWD on wet surface. Since this is the case we chose FWD over RWD in task 3.3 as we reach full traction in lesser time as well.

Power consumption of vehicles without TC

The figures given below with the power consumption without TC when we compare it with the power consumption TC for wet surface RWD. Since there is no traction control and it's slipping alot alot of the torque is given instantaneously just having a high slip ratio (not locking) thereby reaching the peak time in almost no time. Hence this is why the peak power is reached with very little transient time. In the ideal case with TC enabled the wheel slippage is controlled thus the torques are gradually increased.

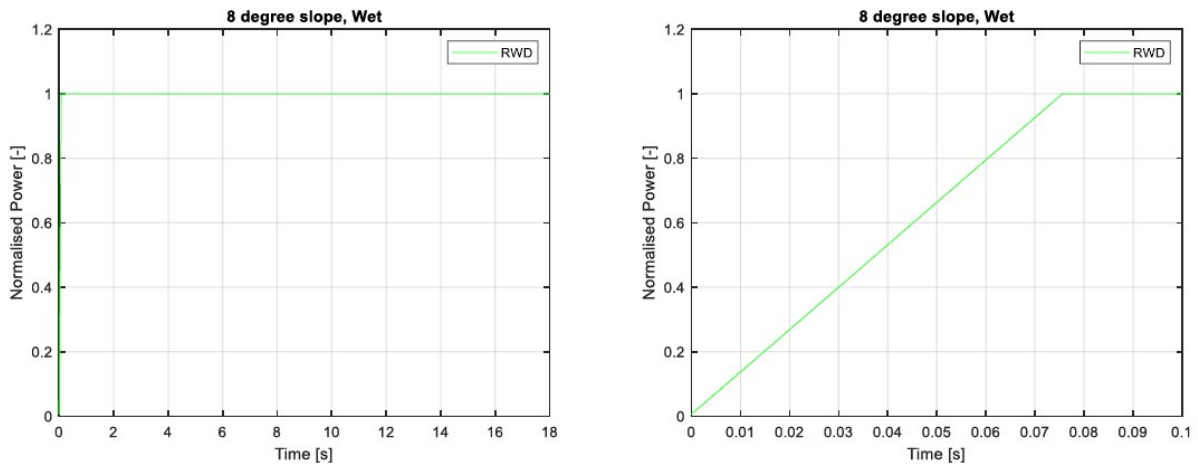


Figure 17: Normilised power vs Time